Calculus Homework 02.11.2026

Zadanie	1	2	3	4	Suma
Punkty					

Zadanie 1:

19. Suppose that $x^2 + y^2 = t$ and that $x = 3$, $y = 4$, and $dx/dt = 7$ when $t = 25$. What is dy/dt at that moment?

Zadanie 2:

23. Water is being pumped from a 20-meter square pond into a round pond with radius 10 meters. At a certain moment, the water level in the square pond is dropping by 2 inches per minute. How fast is the water rising in the round pond?

Zadanie 3:

25. A point in the plane moves in such a way that it is always twice as far from $(0,0)$ as it is from $(0,1)$.

(a) Show that the point moves on a circle.

(b) At the moment when the point crosses the segment between $(0,0)$ and $(0,1)$, what is dy/dt?

(c) Where is the point when $dy/dt = dx/dt$? (You may assume that dx/dt and dy/dt are not simultaneously zero.)

Zadanie 4:

31. (a) Give a rule for determining when the tangent line to a parametric curve $x = f(t)$, $y = g(t)$ is horizontal and when it is vertical.

(b) When is the tangent line to the curve $x = t^2$, $y = t^3 - t$ horizontal? When is it vertical?